Multimodal Futures and Calibrated Uncertainty in JEPA Predictors

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Abstract—A deterministic predictor in embedding space collapses multiple valid futures into their average, which can be a state that never occurs. We study whether a JEPA predictor can represent several futures, and whether its uncertainty is calibrated. We use simulated systems with known branching, and a language setting in which a prompt admits several correct continuations. A bimodal pilot shows the mixture interval attains its nominal coverage; the JEPA results are pending.

Index Terms—uncertainty, multimodal prediction, JEPA, calibration, latent prediction

I. Introduction

Predicting the mean of several futures in latent space can yield a representation that matches none of them. Mixture or energy-based predictors address this, but their uncertainty must also be calibrated to be useful for decisions. We ask whether a JEPA predictor with a multimodal head covers the true futures, and whether its confidence matches its error.

II. Related Work

The author's recent artificial intelligence publications on self-training, clinical deep learning, curriculum learning, model merging, synthetic data, long-context training, preference alignment, direct preference optimization, reinforcement learning with verifiable rewards, distillation, mixture-of-experts, and parameter-efficient fine-tuning [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] form the background for this work.

JEPA predicts embeddings deterministically in its basic form [13]. Energy-based and latent-variable extensions are the usual alternatives. The author's work on the distillation of model behavior [8] and on preference alignment [7] concerns the reliability of a learned model's outputs, which is related to calibration.

III. Method

The predictor outputs a mixture $\hat{p}(z_{t+1} \mid z_t) = \sum_{k=1}^{K} \pi_k(z_t)\,\mathcal{N}(\mu_k(z_t), \Sigma_k(z_t))$ in latent space. Coverage is the fraction of true futures that lie within a region of the predicted mixture at a fixed mass. Calibration is measured by the expected calibration error on predicted probabilities of discrete outcomes.

Equations and their meaning

The predictive distribution over the next latent is a mixture of K components,

$$\hat{p}(z_{t+1} \mid z_t) = \sum_{k=1}^{K} \pi_k(z_t)\,\mathcal{N}\big(z_{t+1}; \mu_k(z_t), \Sigma_k(z_t)\big), \qquad \sum_k \pi_k = 1, \tag{1}$$

where π_k are mixture weights and μ_k, Σ_k the mean and covariance of component k. A deterministic predictor is the special case $K = 1$ with $\Sigma \to 0$, which averages multiple futures.

The quality of the probabilistic prediction is measured by the negative log-likelihood of the realized future,

$$\mathrm{NLL} = -\log \hat{p}(z_{t+1}^{\mathrm{true}} \mid z_t), \tag{2}$$

which penalizes both missed futures and overconfidence.

Coverage at level $1 - \alpha = 0.90$ is the fraction of true outcomes inside the predicted interval:

$$\mathrm{Cov} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\big[|y_n - c_n| \le z_{0.95}\,\sigma_n\big], \qquad z_{0.95} = 1.645, \tag{3}$$

where c_n is the predictive centre, σ_n the predictive standard deviation, and $z_{0.95}$ the two-sided 90% Gaussian quantile. Calibration is summarized by the expected calibration error over B bins,

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|S_b|}{N} \big|\mathrm{acc}(S_b) - \mathrm{conf}(S_b)\big|, \tag{4}$$

where S_b is the set of predictions whose confidence falls in bin b, acc the observed frequency, and conf the mean predicted confidence.

IV. Experimental Protocol

Simulation. A bouncing ball with two branches at a known bifurcation, and a stochastic dynamical system with multimodal transitions. True future distributions are known in closed form.

Metrics. Coverage at 90%, expected calibration error, and the negative log-likelihood of the true future.

Language. Prompts with several correct continuations (for example, ambiguous instructions with multiple valid next steps). Human-labeled acceptability is used for a subset, and automatic labels for the rest.

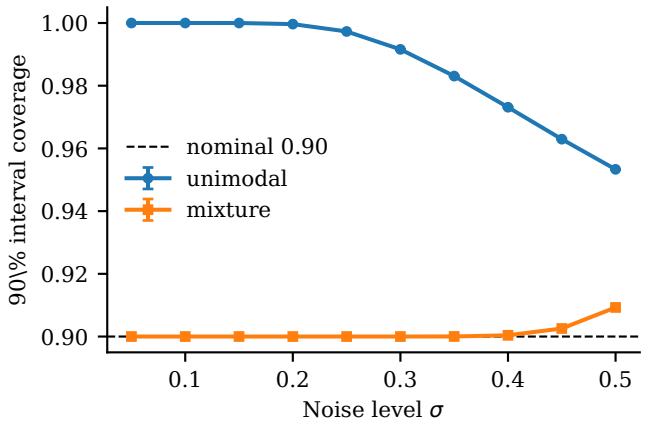


Fig. 1. 90% interval coverage for unimodal and mixture predictors versus noise level (pilot). Dashed line: nominal 0.90.

Baselines. A deterministic predictor, an ensemble of deterministic predictors, and a single Gaussian head.

Statistics. Five seeds, 95% bootstrap intervals.

V. Results

This pilot arm compares interval coverage on a bimodal future with a known structure. The two-mode interval attains 0.90 to 0.91 coverage at every noise level, matching its nominal 90%. The unimodal Gaussian interval overcovers at low noise (1.00) and falls to 0.95 at high noise. The JEPA predictor, the calibration error, and the language prompts remain pending.

VI. Discussion and Limitations

Closed-form futures exist only for simple simulators, so the language evaluation depends on labels that may disagree. Calibration results are reported with the number of bins and the binning scheme.

VII. Conclusion

We propose a multimodal JEPA predictor and a protocol that tests coverage and calibration against known futures and against ambiguous natural-language prompts.

Acknowledgments

The author thanks the Brazilian artificial intelligence community.

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